

Second-Order Sufficient Conditions

Learning objectives

After studying this chapter, you should be able to

- state and prove the one-dimensional and multivariate second-order sufficient conditions;
- explain why positive definiteness gives a uniform local curvature margin and local quadratic growth;
- prove the indefinite-Hessian saddle test;
- distinguish what a local second-order certificate does and does not imply;
- classify stationary points by combining stationarity with Hessian definiteness, and use curved paths when the Hessian is singular; and
- derive and apply the complete minimization classification of an unconstrained quadratic function.

Chapter 6 established necessary conditions for a smooth optimizer. On an open domain, a local minimum must satisfy

$$\nabla f(\mathbf{x}^*) = \mathbf{0}, \quad \mathbf{H}_f(\mathbf{x}^*) \succeq 0,$$

but a singular positive-semidefinite Hessian can hide decisive higher-order terms. The natural question is whether strengthening semidefiniteness to positive definiteness turns the one-way implication into a local certificate.

The global convex certificate has already been proved in [Corollary 4.6](#); it is not repeated here. This chapter assumes no convexity and develops a different conclusion: stationarity together with a definite Hessian certifies a *strict local* optimizer. Quadratic functions are then studied as exact Taylor models with no higher-order remainder. Their complete classification will also be used later in the duality chapters.

7.1 Second-order sufficient condition: one-dimensional cases

At a stationary point x^* , the second-order Taylor expansion is

$$\phi(x^* + h) - \phi(x^*) = \frac{1}{2}\phi''(x^*)h^2 + o(h^2).$$

If $\phi''(x^*)$ is strictly positive, the quadratic term has a positive margin that dominates the remainder for all sufficiently small nonzero h .

Theorem 7.1 (One-dimensional second-order sufficient condition). *Let $\mathcal{D} \subseteq \mathbb{R}$ be open, let $x^* \in \mathcal{D}$, and let $\phi \in C^2(\mathcal{D})$.*

- (a) If $\phi'(x^*) = 0$ and $\phi''(x^*) > 0$, then x^* is a strict local minimizer.
 (b) If $\phi'(x^*) = 0$ and $\phi''(x^*) < 0$, then x^* is a strict local maximizer.

Proof. We prove part (a). Set $m := \phi''(x^*) > 0$. Taylor expansion at x^* , together with stationarity, gives

$$\phi(x^* + h) - \phi(x^*) = \frac{1}{2}mh^2 + r(h), \quad \frac{r(h)}{h^2} \rightarrow 0.$$

Choose $\delta > 0$ so that $x^* + h \in \mathcal{D}$ and $|r(h)| \leq mh^2/4$ whenever $0 < |h| < \delta$. Then

$$\phi(x^* + h) - \phi(x^*) \geq \frac{m}{4}h^2 > 0,$$

so x^* is a strict local minimizer. Part (b) follows by applying part (a) to $-\phi$. $\square$

Example 7.1 (A local certificate is not a global certificate). Define $\phi : \mathbb{R} \rightarrow \mathbb{R}$ by

$$\phi(x) := x^2 + x^3.$$

At the origin, $\phi'(0) = 0$ and $\phi''(0) = 2 > 0$, so the theorem gives a strict local minimum. Directly, for $0 < |x| < 1$,

$$\phi(x) = x^2(1 + x) > 0 = \phi(0).$$

Nevertheless, $\phi(x) \rightarrow -\infty$ as $x \rightarrow -\infty$. Thus the theorem does not imply a global minimum.

The same local certificate also does not identify a global minimizer when one exists. Indeed, let

$$\psi(x) := x^2 - 3x^4 + x^6.$$

Again $\psi'(0) = 0$ and $\psi''(0) = 2 > 0$, so the origin is a strict local minimizer. The function is coercive and therefore has a global minimizer, but

$$\psi(1) = -1 < 0 = \psi(0).$$

Consequently, the origin is not a global minimizer. The second-order sufficient condition is purely local: by itself it decides neither global existence nor global optimality.

7.2 Second-order sufficient condition: multi-dimensional cases

For a symmetric positive-definite matrix $\mathbf{H}$, compactness of the unit sphere gives the uniform lower bound

$$\mathbf{d}^\top \mathbf{H} \mathbf{d} \geq \lambda_{\min}(\mathbf{H}) \|\mathbf{d}\|^2 \quad (\mathbf{d} \in \mathbb{R}^n), \quad (7.1)$$

where $\lambda_{\min}(\mathbf{H}) > 0$. This single constant controls every direction simultaneously.

Theorem 7.2 (Second-order sufficient condition and quadratic growth). *Let $\mathcal{D} \subseteq \mathbb{R}^n$ be open, let $\mathbf{x}^* \in \mathcal{D}$, and let $f \in C^2(\mathcal{D})$.*

- (a) *If $\nabla f(\mathbf{x}^*) = \mathbf{0}$ and $\mathbf{H}_f(\mathbf{x}^*) \succ 0$, then there exist constants $c > 0$ and $r > 0$, with $B(\mathbf{x}^*, r) \subseteq \mathcal{D}$, such that*

$$f(\mathbf{x}) \geq f(\mathbf{x}^*) + c \|\mathbf{x} - \mathbf{x}^*\|^2 \quad \text{for every } \mathbf{x} \in B(\mathbf{x}^*, r). \quad (7.2)$$

In particular, $\mathbf{x}^$ is a strict local minimizer.*

(b) If $\nabla f(\mathbf{x}^) = \mathbf{0}$ and $\mathbf{H}_f(\mathbf{x}^*) \prec 0$, then $\mathbf{x}^*$ is a strict local maximizer.*

Proof. We prove part (a). Let

$$m := \lambda_{\min}(\mathbf{H}_f(\mathbf{x}^*)) > 0.$$

Continuity of the Hessian in the operator norm gives $r > 0$ such that $B(\mathbf{x}^*, r) \subseteq \mathcal{D}$ and

$$\|\mathbf{H}_f(\mathbf{x}) - \mathbf{H}_f(\mathbf{x}^*)\|_2 \leq \frac{m}{2} \quad (\mathbf{x} \in B(\mathbf{x}^*, r)).$$

Consequently, for every $\mathbf{d} \in \mathbb{R}^n$ and every $\mathbf{x} \in B(\mathbf{x}^*, r)$,

$$\begin{aligned} \mathbf{d}^\top \mathbf{H}_f(\mathbf{x}) \mathbf{d} &\geq \mathbf{d}^\top \mathbf{H}_f(\mathbf{x}^*) \mathbf{d} - \|\mathbf{H}_f(\mathbf{x}) - \mathbf{H}_f(\mathbf{x}^*)\|_2 \|\mathbf{d}\|^2 \\ &\geq \frac{m}{2} \|\mathbf{d}\|^2. \end{aligned}$$

Fix a nonzero $\mathbf{d}$ with $\|\mathbf{d}\| < r$, and define $\psi(t) := f(\mathbf{x}^* + t\mathbf{d})$ for $0 \leq t \leq 1$. Twice applying the fundamental theorem of calculus gives

$$\psi(1) - \psi(0) = \psi'(0) + \int_0^1 (1-t)\psi''(t) dt.$$

Since $\psi'(0) = \nabla f(\mathbf{x}^*)^\top \mathbf{d} = 0$ and $\psi''(t) = \mathbf{d}^\top \mathbf{H}_f(\mathbf{x}^* + t\mathbf{d}) \mathbf{d}$,

$$f(\mathbf{x}^* + \mathbf{d}) - f(\mathbf{x}^*) \geq \int_0^1 (1-t) \frac{m}{2} \|\mathbf{d}\|^2 dt = \frac{m}{4} \|\mathbf{d}\|^2.$$

Thus Equation (7.2) holds with $c = m/4$, proving part (a). Applying part (a) to $-f$ proves part (b). $\square$

The quadratic-growth conclusion is quantitatively stronger than strict local minimality: it supplies the explicit lower bound $f(\mathbf{x}) - f(\mathbf{x}^*) \geq c \|\mathbf{x} - \mathbf{x}^*\|^2$, rather than merely asserting that the difference is positive.

Corollary 7.3 (Indefinite Hessian test). *Let $\mathcal{D} \subseteq \mathbb{R}^n$ be open, let $\mathbf{x}^* \in \mathcal{D}$, and let $f \in C^2(\mathcal{D})$. If $\mathbf{x}^*$ is stationary and $\mathbf{H}_f(\mathbf{x}^*)$ is indefinite, then $\mathbf{x}^*$ is a saddle point.*

Proof. Since $\mathbf{H}_f(\mathbf{x}^*)$ is indefinite, there exist directions $\mathbf{d}_+, \mathbf{d}_- \in \mathbb{R}^n$ such that

$$\mathbf{d}_+^\top \mathbf{H}_f(\mathbf{x}^*) \mathbf{d}_+ > 0, \quad \mathbf{d}_-^\top \mathbf{H}_f(\mathbf{x}^*) \mathbf{d}_- < 0.$$

For $\mathbf{d} \in \{\mathbf{d}_+, \mathbf{d}_-\}$, Taylor expansion at the stationary point $\mathbf{x}^*$ gives

$$f(\mathbf{x}^* + t\mathbf{d}) - f(\mathbf{x}^*) = \frac{t^2}{2} \mathbf{d}^\top \mathbf{H}_f(\mathbf{x}^*) \mathbf{d} + o(t^2) \quad (t \rightarrow 0).$$

For all sufficiently small nonzero t , this difference is positive when $\mathbf{d} = \mathbf{d}_+$ and negative when $\mathbf{d} = \mathbf{d}_-$. Thus every neighbourhood of $\mathbf{x}^*$ contains points with objective values both above and below $f(\mathbf{x}^*)$. Hence $\mathbf{x}^*$ is neither a local minimizer nor a local maximizer; since it is stationary, it is a saddle point. $\square$

Warning

Stationarity cannot be omitted. For $\phi(x) = x + x^2$, one has $\phi''(0) = 2 > 0$, but $\phi'(0) = 1$, so 0 is not a local minimizer. The unique global minimizer is $-1/2$. Positive curvature describes the shape of the graph, but a nonzero first-order slope still produces descent.

Warning

Positive definiteness is sufficient but not necessary. Define

$$g(x_1, x_2) := (x_1 + x_2)^2 + x_2^4.$$

At the origin,

$$\nabla g(\mathbf{0}) = \mathbf{0}, \quad \mathbf{H}_g(\mathbf{0}) = 2 \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \succeq 0$$

is singular. Nevertheless, $g \geq 0$, with equality only at the origin, so the origin is the unique global minimizer. Along the kernel direction $\mathbf{d} = (1, -1)^\top$, one has $g(t, -t) = t^4$. Thus the minimum is strict but does not have quadratic growth, since $g(t, -t)/\|(t, -t)\|^2 = t^2/2 \rightarrow 0$. A singular positive-semidefinite Hessian remains inconclusive in general; the opposite outcome appears in [Example 7.3](#).

7.2.1 The decision chart

Combining [Theorem 7.2](#) with the indefinite-Hessian test in [Corollary 7.3](#) gives the following chart for an interior stationary point $\mathbf{x}^*$:

Hessian at $\mathbf{x}^*$	conclusion
positive definite	strict local minimizer
negative definite	strict local maximizer
indefinite	saddle point
singular positive or negative semidefinite	no conclusion in general

The first two rows are sufficient conditions. The third row rejects both a local minimum and a local maximum. In the final row, one must inspect higher-order terms, restrictions to curves, or a direct neighborhood inequality.

Example 7.2 (A coupled two-dimensional classification). Consider

$$F(x_1, x_2) := 2x_1^3 + 6x_1x_2^2 - 6x_1^2 - 6x_2^2 + 2.$$

Unlike a coordinatewise sum, this polynomial contains the coupling term $x_1x_2^2$. Its gradient and Hessian are

$$\nabla F(x_1, x_2) = \begin{bmatrix} 6(x_1^2 + x_2^2 - 2x_1) \\ 12x_2(x_1 - 1) \end{bmatrix}, \quad \mathbf{H}_F(x_1, x_2) = 12 \begin{bmatrix} x_1 - 1 & x_2 \\ x_2 & x_1 - 1 \end{bmatrix}.$$

We first solve the stationary equations exhaustively. The second equation gives $x_2 = 0$ or $x_1 = 1$. If $x_2 = 0$, the first equation becomes

$$x_1(x_1 - 2) = 0,$$

and hence gives $(0, 0)$ and $(2, 0)$. If $x_1 = 1$, the first equation becomes $x_2^2 = 1$, giving $(1, 1)$ and $(1, -1)$. Therefore the full stationary set is

$$\mathcal{S} = \{(0, 0), (2, 0), (1, 1), (1, -1)\}.$$

At the first two points,

$$\mathbf{H}_F(0, 0) = -12I_2 \prec 0, \quad \mathbf{H}_F(2, 0) = 12I_2 \succ 0.$$

Thus $(0, 0)$ is a strict local maximizer and $(2, 0)$ is a strict local minimizer by [Theorem 7.2](#). For $\sigma \in \{1, -1\}$,

$$\mathbf{H}_F(1, \sigma) = 12 \begin{bmatrix} 0 & \sigma \\ \sigma & 0 \end{bmatrix}$$

has eigenvalues 12 and -12 . It is therefore indefinite, so both $(1, 1)$ and $(1, -1)$ are saddle points by [Corollary 7.3](#).

These conclusions are local. Indeed, along $x_2 = 0$,

$$F(t, 0) = 2t^3 - 6t^2 + 2,$$

which tends to $-\infty$ as $t \rightarrow -\infty$ and to $+\infty$ as $t \rightarrow +\infty$. Hence F has neither a global minimum nor a global maximum.

Example 7.3 (A singular Hessian that requires a curved path). Consider

$$f(x_1, x_2) := (x_1 + x_2^2)^2 - x_2^6.$$

Its gradient is

$$\nabla f(x_1, x_2) = \begin{bmatrix} 2(x_1 + x_2^2) \\ 4x_2(x_1 + x_2^2) - 6x_2^5 \end{bmatrix},$$

and hence $\nabla f(\mathbf{0}) = \mathbf{0}$. The Hessian is

$$\mathbf{H}_f(x_1, x_2) = \begin{bmatrix} 2 & 4x_2 \\ 4x_2 & 4x_1 + 12x_2^2 - 30x_2^4 \end{bmatrix}, \quad \mathbf{H}_f(\mathbf{0}) = \text{diag}(2, 0) \succeq 0.$$

Thus the second-order decision chart is inconclusive, and

$$\text{Null}(\mathbf{H}_f(\mathbf{0})) = \text{span}\{(0, 1)^\top\}.$$

Along the kernel line,

$$f(0, t) = t^4(1 - t^2) > 0 \quad (0 < |t| < 1).$$

In fact, every fixed line through the origin initially gives positive values. For a fixed $(a, b) \neq (0, 0)$,

$$f(ta, tb) = t^2(a + tb^2)^2 - t^6b^6.$$

If $a \neq 0$, the leading nonzero term is $a^2t^2 > 0$; if $a = 0$, then $b \neq 0$ and

$$f(0, tb) = t^4b^4(1 - t^2b^2) > 0$$

for every sufficiently small nonzero t .

Nevertheless, the curve $x_1 = -x_2^2$ cancels the nonnegative square. Its parameterization $(x_1, x_2) = (-t^2, t)$ gives

$$f(-t^2, t) = -t^6 < 0 \quad (t \neq 0).$$

Since arbitrarily close points on fixed lines have values above $f(\mathbf{0}) = 0$, whereas arbitrarily close points on this curved path have values below it, the origin is a saddle point.

Warning

When the Hessian is singular, checking the coordinate axes—or even every fixed line through the point—need not classify the point. A descent path may bend toward the Hessian kernel as it approaches the point, as in [Example 7.3](#).

Important note: classification is not computation

The Hessian test classifies a stationary point after it has been found. It does not prove existence of a global optimizer, establish globality, or solve the nonlinear system $\nabla f(\mathbf{x}) = \mathbf{0}$. A singular Hessian can also require direct inequalities or carefully chosen curves.

7.3 Quadratic functions as exact Taylor models

Definition 7.1 (Quadratic function). Let $\mathbf{Q} \in \mathbb{R}^{n \times n}$ be symmetric, let $\mathbf{c} \in \mathbb{R}^n$, and let $\gamma \in \mathbb{R}$. A quadratic function on $\mathbb{R}^n$ has the form

$$q(\mathbf{x}) := \frac{1}{2} \mathbf{x}^\top \mathbf{Q} \mathbf{x} + \mathbf{c}^\top \mathbf{x} + \gamma.$$

The symmetry assumption entails no loss of generality because the skew-symmetric part of a matrix contributes zero to its quadratic form.

Lemma 7.4 (Exact quadratic expansion). For every $\mathbf{x}, \mathbf{d} \in \mathbb{R}^n$,

$$q(\mathbf{x} + \mathbf{d}) - q(\mathbf{x}) = (\mathbf{Q}\mathbf{x} + \mathbf{c})^\top \mathbf{d} + \frac{1}{2} \mathbf{d}^\top \mathbf{Q} \mathbf{d}. \quad (7.3)$$

Consequently,

$$\nabla q(\mathbf{x}) = \mathbf{Q}\mathbf{x} + \mathbf{c}, \quad H_q(\mathbf{x}) = \mathbf{Q}.$$

Proof. Expanding and using $\mathbf{Q}^\top = \mathbf{Q}$,

$$\begin{aligned} q(\mathbf{x} + \mathbf{d}) &= \frac{1}{2} (\mathbf{x} + \mathbf{d})^\top \mathbf{Q} (\mathbf{x} + \mathbf{d}) + \mathbf{c}^\top (\mathbf{x} + \mathbf{d}) + \gamma \\ &= q(\mathbf{x}) + (\mathbf{Q}\mathbf{x} + \mathbf{c})^\top \mathbf{d} + \frac{1}{2} \mathbf{d}^\top \mathbf{Q} \mathbf{d}. \end{aligned}$$

The gradient and Hessian follow by comparing the linear and quadratic terms in $\mathbf{d}$. □

Since the Hessian is the constant matrix $\mathbf{Q}$, the Hessian characterization in [Theorem 5.5](#) immediately gives

$$q \text{ is convex} \iff \mathbf{Q} \succeq 0.$$

It also gives strict convexity when $\mathbf{Q} \succ 0$. Conversely, if $\mathbf{Q} \succeq 0$ is singular, any nonzero $\mathbf{z} \in \text{Null}(\mathbf{Q})$ makes $t \mapsto q(\mathbf{x} + t\mathbf{z})$ affine on every line parallel to $\mathbf{z}$. Hence

$$q \text{ is strictly convex} \iff \mathbf{Q} \succ 0.$$

The next theorem classifies minimization of every unconstrained quadratic, including indefinite and singular cases. It is proved directly and does not repeat the general convex optimality theorem from [Chapter 4](#).

Lemma 7.5 (Range–nullspace orthogonality). *If $\mathbf{Q} \in \mathbb{R}^{n \times n}$ is symmetric, then*

$$\text{Range}(\mathbf{Q}) = \text{Null}(\mathbf{Q})^\perp.$$

Proof. Let $\mathbf{y} \in \text{Range}(\mathbf{Q})$ and $\mathbf{z} \in \text{Null}(\mathbf{Q})$. Write $\mathbf{y} = \mathbf{Q}\mathbf{x}$. Symmetry gives

$$\mathbf{y}^\top \mathbf{z} = \mathbf{x}^\top \mathbf{Q}^\top \mathbf{z} = \mathbf{x}^\top \mathbf{Q} \mathbf{z} = 0.$$

Thus $\text{Range}(\mathbf{Q}) \subseteq \text{Null}(\mathbf{Q})^\perp$. By rank–nullity,

$$\dim \text{Range}(\mathbf{Q}) = n - \dim \text{Null}(\mathbf{Q}) = \dim \text{Null}(\mathbf{Q})^\perp.$$

The inclusion and equality of dimensions yield the result. $\square$

7.4 Complete classification of unconstrained quadratics

Theorem 7.6 (Complete quadratic minimization classification). *Let $\mathbf{Q} \in \mathbb{R}^{n \times n}$ be symmetric, let $\mathbf{c} \in \mathbb{R}^n$, and let $\gamma \in \mathbb{R}$. Define*

$$q(\mathbf{x}) := \frac{1}{2} \mathbf{x}^\top \mathbf{Q} \mathbf{x} + \mathbf{c}^\top \mathbf{x} + \gamma.$$

Exactly one of the following cases applies.

- (i) *If $\mathbf{Q} \not\geq 0$, then $\inf_{\mathbb{R}^n} q = -\infty$.*
- (ii) *If $\mathbf{Q} \succeq 0$ and $\mathbf{c} \notin \text{Range}(\mathbf{Q})$, then $\inf_{\mathbb{R}^n} q = -\infty$.*
- (iii) *If $\mathbf{Q} \succeq 0$ and $\mathbf{c} = \mathbf{Q}\mathbf{w}$, then*

$$\min_{\mathbb{R}^n} q = \gamma - \frac{1}{2} \mathbf{w}^\top \mathbf{Q} \mathbf{w}, \quad \arg \min_{\mathbb{R}^n} q = -\mathbf{w} + \text{Null}(\mathbf{Q}).$$

The value and affine minimizer set in part (iii) do not depend on which solution $\mathbf{w}$ of $\mathbf{Q}\mathbf{w} = \mathbf{c}$ is chosen.

Proof. If $\mathbf{Q} \not\geq 0$, choose $\mathbf{d}$ with $\mathbf{d}^\top \mathbf{Q} \mathbf{d} < 0$. Along the line $t\mathbf{d}$,

$$q(t\mathbf{d}) = \frac{1}{2} t^2 \mathbf{d}^\top \mathbf{Q} \mathbf{d} + t \mathbf{c}^\top \mathbf{d} + \gamma \longrightarrow -\infty \quad (|t| \rightarrow \infty),$$

because the negative quadratic term dominates the linear term. This proves part (i).

Suppose next that $\mathbf{Q} \succeq 0$ and $\mathbf{c} \notin \text{Range}(\mathbf{Q})$. By [Lemma 7.5](#), there exists $\mathbf{z} \in \text{Null}(\mathbf{Q})$ with $\mathbf{c}^\top \mathbf{z} \neq 0$. Replacing $\mathbf{z}$ by $-\mathbf{z}$ if necessary, assume $\mathbf{c}^\top \mathbf{z} < 0$. Then

$$q(t\mathbf{z}) = t \mathbf{c}^\top \mathbf{z} + \gamma \longrightarrow -\infty \quad (t \rightarrow +\infty),$$

which proves part (ii).

Finally, suppose $\mathbf{Q} \succeq 0$ and $\mathbf{c} = \mathbf{Q}\mathbf{w}$. Completing the square gives

$$q(\mathbf{x}) = \frac{1}{2}(\mathbf{x} + \mathbf{w})^\top \mathbf{Q}(\mathbf{x} + \mathbf{w}) + \gamma - \frac{1}{2}\mathbf{w}^\top \mathbf{Q}\mathbf{w}.$$

The first term is nonnegative and vanishes exactly when $\mathbf{x} + \mathbf{w} \in \text{Null}(\mathbf{Q})$, proving the value and minimizer set.

If $\mathbf{w}_1$ and $\mathbf{w}_2$ both satisfy $\mathbf{Q}\mathbf{w}_i = \mathbf{c}$, then $\mathbf{w}_1 - \mathbf{w}_2 \in \text{Null}(\mathbf{Q})$. Hence $-\mathbf{w}_1 + \text{Null}(\mathbf{Q}) = -\mathbf{w}_2 + \text{Null}(\mathbf{Q})$. Moreover,

$$\mathbf{w}_1^\top \mathbf{Q}\mathbf{w}_1 - \mathbf{w}_2^\top \mathbf{Q}\mathbf{w}_2 = (\mathbf{w}_1 - \mathbf{w}_2)^\top \mathbf{c} = 0,$$

so the minimum value is also independent of the choice. $\square$

Corollary 7.7 (Positive-definite quadratic). *If $\mathbf{Q} \succ 0$, then q has the unique global minimizer*

$$\mathbf{x}^* = -\mathbf{Q}^{-1}\mathbf{c}.$$

For every $\mathbf{x} \neq \mathbf{x}^*$,

$$q(\mathbf{x}) - q(\mathbf{x}^*) = \frac{1}{2}(\mathbf{x} - \mathbf{x}^*)^\top \mathbf{Q}(\mathbf{x} - \mathbf{x}^*) > 0.$$

Proof. Positive definiteness makes $\mathbf{Q}$ invertible and $\text{Null}(\mathbf{Q}) = \{\mathbf{0}\}$. Apply part (iii) of [Theorem 7.6](#) with $\mathbf{w} = \mathbf{Q}^{-1}\mathbf{c}$. The displayed identity follows from [Equation \(7.3\)](#) at the stationary point. $\square$

Warning

Positive semidefiniteness of $\mathbf{Q}$ does not by itself guarantee a minimizer. With $\mathbf{Q} = \text{diag}(1, 0)$, the function $q_1(\mathbf{x}) = \frac{1}{2}x_1^2$ has the whole line $x_1 = 0$ as minimizers, whereas $q_2(\mathbf{x}) = \frac{1}{2}x_1^2 + x_2$ is unbounded below. The linear coefficient along the nullspace decides between these outcomes.

Warning

A stationary point of an indefinite quadratic is not a minimizer. For $q(x_1, x_2) = x_1^2 - x_2^2$, the origin is stationary, but the Hessian is indefinite. The function is positive on the x_1 -axis, negative on the x_2 -axis, and unbounded below.

Example 7.4 (A positive-definite quadratic). Consider

$$q(x_1, x_2) := x_1^2 + x_1x_2 + 2x_2^2 - 2x_1 - 3x_2.$$

Then

$$\mathbf{Q} = \begin{bmatrix} 2 & 1 \\ 1 & 4 \end{bmatrix} \succ 0, \quad \mathbf{c} = \begin{bmatrix} -2 \\ -3 \end{bmatrix}.$$

Solving $\mathbf{Q}\mathbf{x}^* = -\mathbf{c}$ gives

$$\mathbf{x}^* = \begin{bmatrix} 5/7 \\ 4/7 \end{bmatrix}, \quad q(\mathbf{x}^*) = -\frac{11}{7}.$$

By [Corollary 7.7](#), this is the unique global minimizer.

Example 7.5 (One singular matrix, two outcomes). Let

$$\mathbf{Q} := \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \quad \text{Range}(\mathbf{Q}) = \text{span}\{(1, 1)^\top\}.$$

For $i \in \{1, 2\}$, define

$$q_i(\mathbf{x}) := \frac{1}{2} \mathbf{x}^\top \mathbf{Q} \mathbf{x} + \mathbf{c}_i^\top \mathbf{x},$$

so the constant term is zero. For $\mathbf{c}_1 = (-2, -2)^\top$, one may choose $\mathbf{w} = (-1, -1)^\top$, so the minimum is -2 and

$$\arg \min q_1 = (1, 1)^\top + \text{Null}(\mathbf{Q}) = \{\mathbf{x} : x_1 + x_2 = 2\}.$$

For $\mathbf{c}_2 = (-2, -1)^\top$, the range condition fails, so $\inf q_2 = -\infty$.

7.5 From certificates to computation

At a current point $\mathbf{x}$, the second-order Taylor model of a smooth function is

$$m_{\mathbf{x}}(\mathbf{d}) := f(\mathbf{x}) + \nabla f(\mathbf{x})^\top \mathbf{d} + \frac{1}{2} \mathbf{d}^\top \mathbf{H}_f(\mathbf{x}) \mathbf{d}.$$

It is a quadratic in the displacement $\mathbf{d}$.

Proposition 7.8 (Minimizing a positive-definite local model). *If $\mathbf{H}_f(\mathbf{x}) \succ 0$, then the Taylor model has the unique minimizer*

$$\mathbf{H}_f(\mathbf{x}) \mathbf{d}_N = -\nabla f(\mathbf{x}), \quad \mathbf{d}_N = -\mathbf{H}_f(\mathbf{x})^{-1} \nabla f(\mathbf{x}).$$

In one dimension this becomes

$$d_N = -\frac{f'(x)}{f''(x)}.$$

Proof. The gradient of $m_{\mathbf{x}}$ with respect to $\mathbf{d}$ is

$$\nabla_{\mathbf{d}} m_{\mathbf{x}}(\mathbf{d}) = \nabla f(\mathbf{x}) + \mathbf{H}_f(\mathbf{x}) \mathbf{d}.$$

The Hessian of the model is the positive-definite matrix $\mathbf{H}_f(\mathbf{x})$. Solving the stationary equation gives the displayed unique minimizer by [Corollary 7.7](#). $\square$

In computation, one solves the displayed linear system for $\mathbf{d}_N$; one does not form the inverse matrix explicitly.

This formula foreshadows Newton's method. [Chapter 8](#) first studies how to locate one-dimensional candidates when $f'(x) = 0$ has no closed-form solution; later we return to multivariate quadratic models.

Chapter summary

- At a stationary point, a positive-definite Hessian guarantees a strict local minimum and local quadratic growth; a negative-definite Hessian gives the corresponding maximum statement.
- The sufficient condition is not necessary and does not imply global optimality. A singular semidefinite Hessian is inconclusive, and a curved path may reveal behavior that every fixed line misses.
- An indefinite Hessian at a stationary point gives a saddle point.
- Quadratic functions have an exact second-order expansion. Their minimization is completely determined by the curvature of $\mathbf{Q}$ and whether $\mathbf{c}$ lies in $\text{Range}(\mathbf{Q})$.
- A certificate classifies a candidate; numerical methods are needed when the stationary equation cannot be solved analytically.

7.6 Exercises

Exercise 7.1 (One-dimensional second-order classification). Let

$$\phi(x) := \frac{x}{1+x^2}.$$

Find every stationary point and classify it by the one-dimensional second-order sufficient condition. Then use $2|x| \leq 1+x^2$ to decide whether either point is a global minimum or maximum.

Exercise 7.2 (Intersecting sets of global minimizers). Consider

$$r(x_1, x_2) = x_1^2 x_2^2.$$

Compute $\nabla r(\mathbf{0})$ and $\mathbf{H}_r(\mathbf{0})$, and explain why the second-order chart is inconclusive. Find the complete set of global minimizers. Is the origin a strict local minimizer? Does r have quadratic growth at the origin?

Exercise 7.3 (Three logical boundaries). In each item, the displayed point is an interior stationary point. Decide whether the stated implication is true in general. If it is false, verify that the suggested polynomial is a counterexample.

- If $\mathbf{x}^*$ is a strict local minimizer, then $\mathbf{H}_f(\mathbf{x}^*) \succ 0$. Consider $u(x) = x^8$ at 0 .
- If $\mathbf{H}_f(\mathbf{x}^*) \succeq 0$, then $\mathbf{x}^*$ is a local minimizer. Consider $v(x_1, x_2) = x_1^2 x_2$ at $\mathbf{0}$.
- If $\mathbf{H}_f(\mathbf{x}^*) \succ 0$, then $\mathbf{x}^*$ is a global minimizer. Consider $w(x) = x^2 - 3x^3 + x^4$ at 0 .

Exercise 7.4 (A grid of stationary points). Consider

$$f(x_1, x_2) := (x_1^2 - 4)^2 + (x_2^2 - 1)^2.$$

Find every stationary point and classify it as a strict local minimum, strict local maximum, or saddle point. Determine which local minima are global.

Exercise 7.5 (Fixed lines can miss a curved descent path). Consider

$$g(x_1, x_2) := x_1^2 - 3x_1x_2^3 + x_2^6.$$

- (a) Compute $\nabla g(\mathbf{0})$, $\mathbf{H}_g(\mathbf{0})$, and the nullspace of the Hessian.
- (b) Prove that, on every fixed nonzero line $(x_1, x_2) = t(a, b)$, one has $g(ta, tb) > 0$ for all sufficiently small nonzero t .
- (c) Evaluate $g(x_2^3, x_2)$, and use this curved path to classify the origin.

Exercise 7.6 (A least-squares quadratic). Let

$$\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \quad q(\mathbf{x}) := \frac{1}{2} \|\mathbf{Ax} - \mathbf{b}\|^2.$$

Find $\mathbf{Q}, \mathbf{c}, \gamma$, verify $\mathbf{Q} \succ 0$, and compute $\arg \min q$ and $\min q$.

Exercise 7.7 (A parameterized quadratic). Let $c \in \mathbb{R}$ and

$$q_c(x_1, x_2) := cx_1^2 + x_2^2 - 2x_1x_2 - 2x_2.$$

- (a) Determine the definiteness of its quadratic matrix for every value of c .
- (b) Determine exactly for which values of c a global minimizer exists.
- (c) In every case where a minimum exists, find the complete minimizer set and minimum value. In every remaining case, give a direction along which the objective tends to $-\infty$.

Exercise 7.8 (A singular quadratic). Let

$$\mathbf{Q} = \begin{bmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix}, \quad \mathbf{c} \in \mathbb{R}^3, \quad q(\mathbf{x}) = \frac{1}{2} \mathbf{x}^\top \mathbf{Q} \mathbf{x} + \mathbf{c}^\top \mathbf{x}.$$

- (a) Show that

$$\mathbf{x}^\top \mathbf{Q} \mathbf{x} = (x_1 - x_2)^2 + (x_2 - x_3)^2,$$
 and find $\text{Null}(\mathbf{Q})$ and $\text{Range}(\mathbf{Q})$.
- (b) Determine exactly for which $\mathbf{c}$ a global minimum exists.
- (c) When a minimum exists, find the complete minimizer set and the minimum value in terms of $\mathbf{c}$. Otherwise, exhibit an explicit escape direction.

Exercise 7.9 (Integrated three-variable classification). Define

$$f(x_1, x_2, x_3) := 2x_1x_2x_3 - 4x_1x_3 - 2x_2x_3 + x_1^2 + x_2^2 + x_3^2 - 2x_1 - 4x_2 + 4x_3.$$

Find every stationary point and classify it as a strict local minimum, strict local maximum, or saddle point. To simplify the stationary system, set $a = x_1 - 1$ and $b = x_2 - 2$; it then

becomes

$$a + x_3b = 0, \quad b + x_3a = 0, \quad x_3 + ab = 0.$$